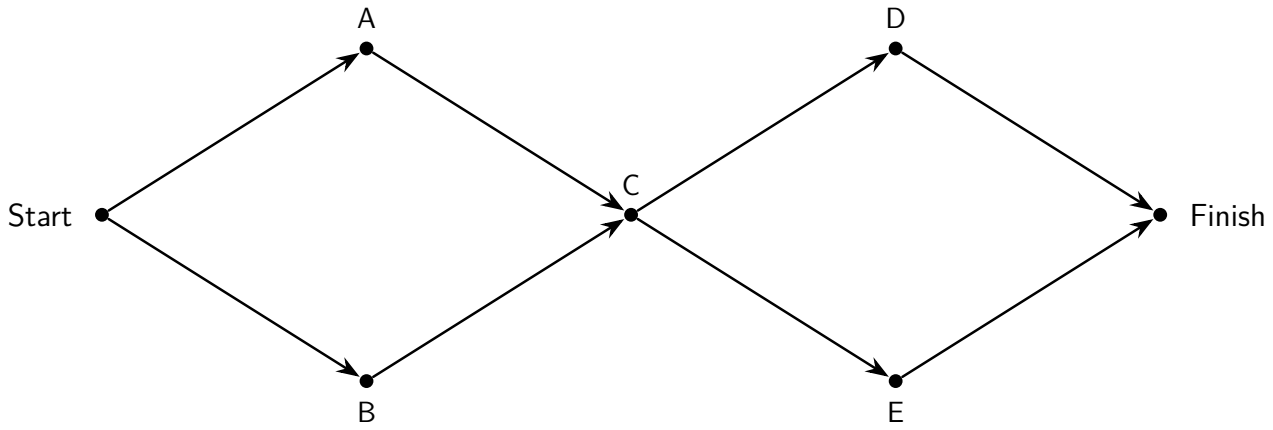
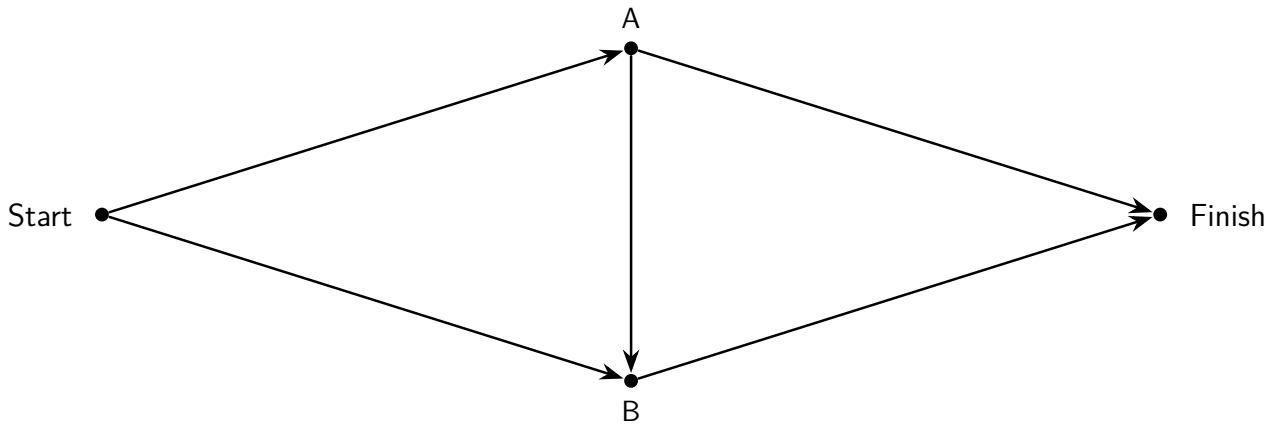
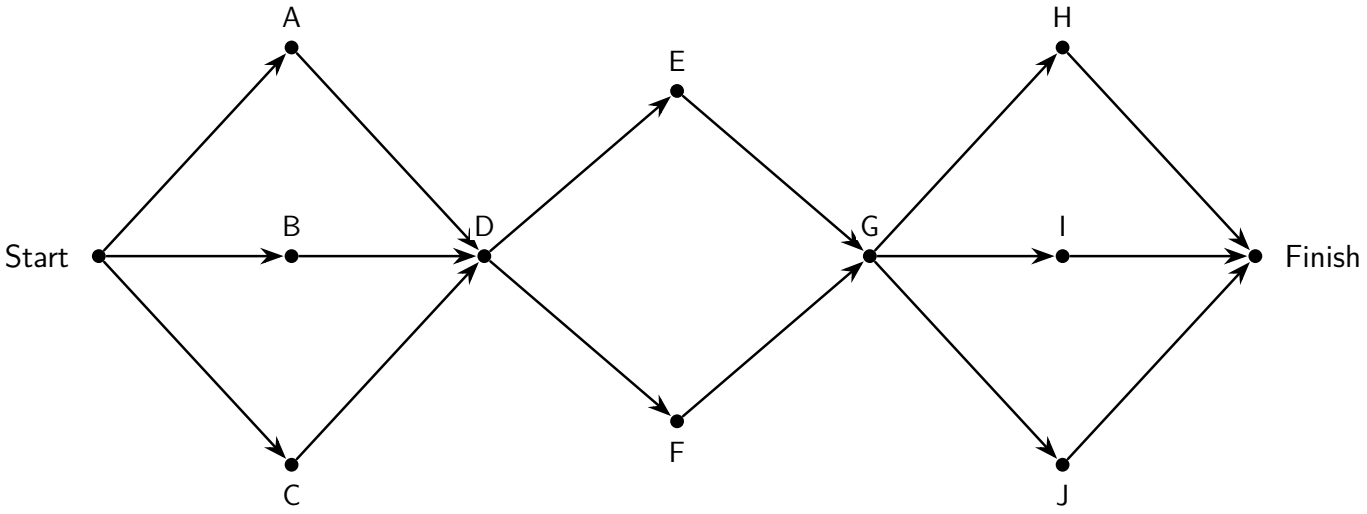
A route follows arrows from Start to Finish without visiting a dot twice. Routes may meet at dots, but each arrow can belong to only one route in a collection. Use a different color or line pattern for each route. A marker closes one arrow.

Problem 1: On each board, fit as many routes as possible. Find the fewest arrows you can close to stop every route.

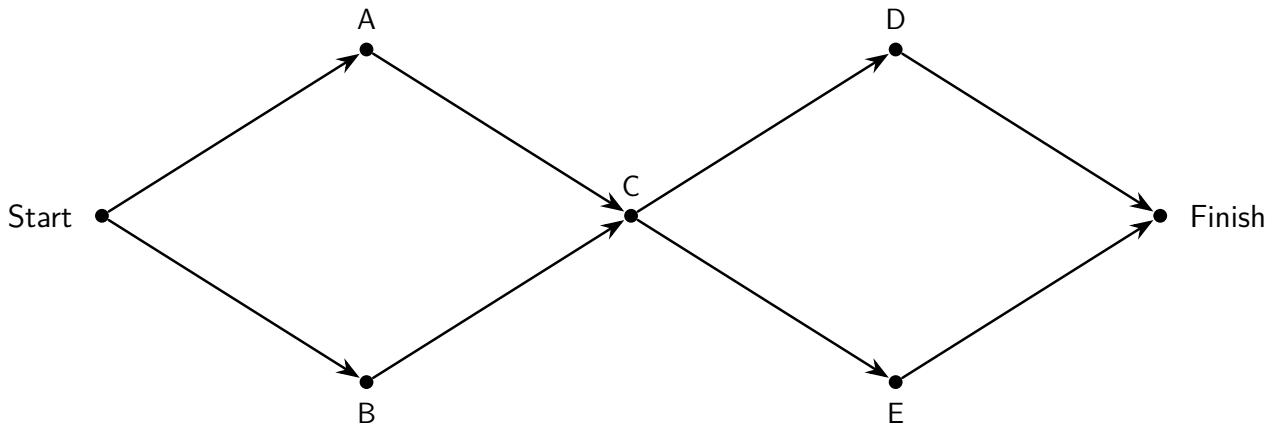




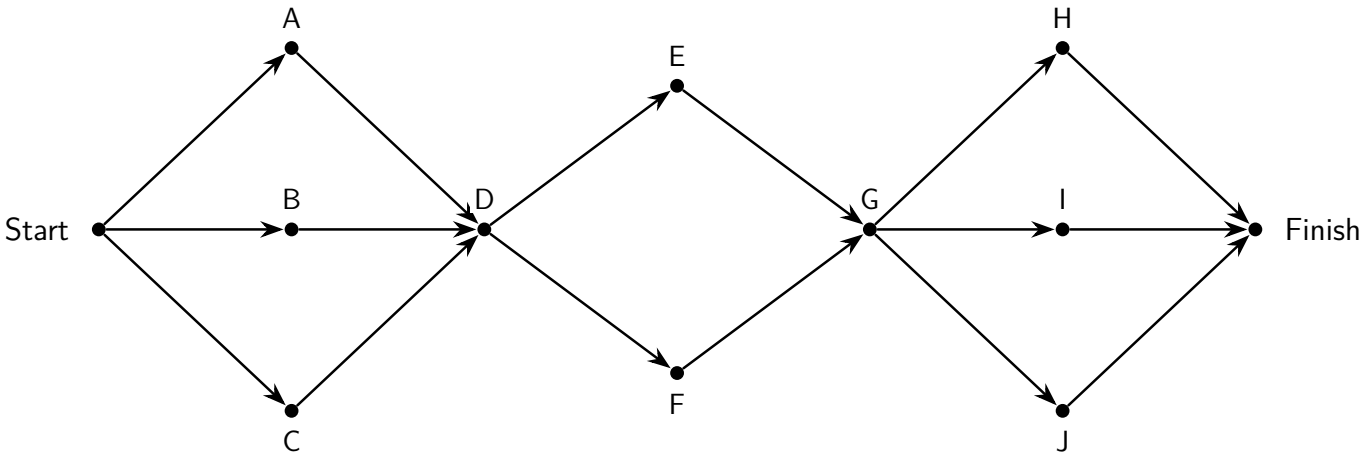
Problem 2: Find the largest collection of routes on this board. Close as few arrows as possible to stop every route. Explain how you know your collection cannot be larger.

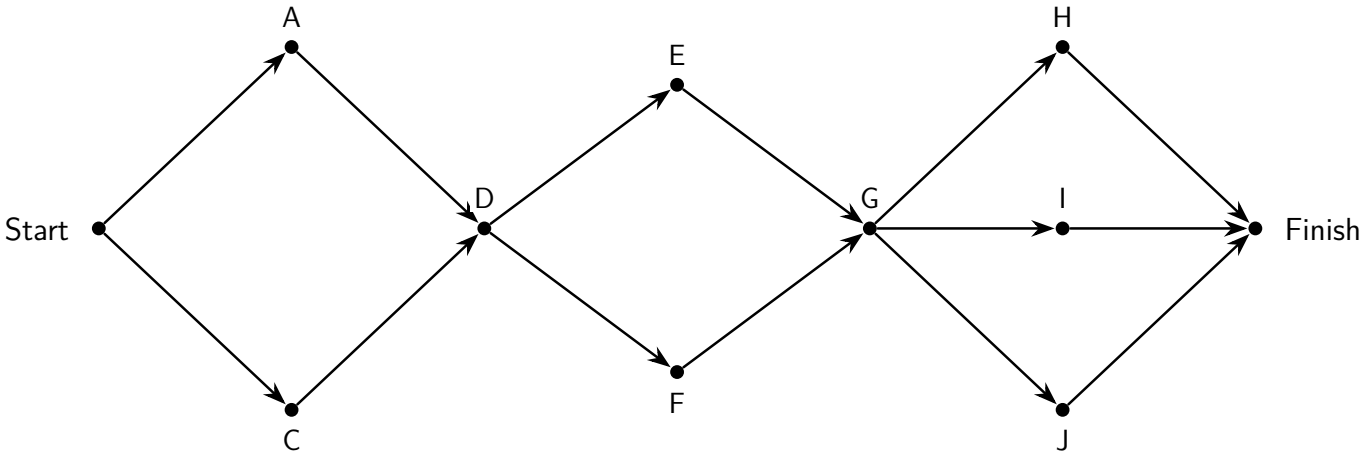


Problem 3: Find every pair of arrows you can close to stop all travel on this board. Explain how you know your list is complete.

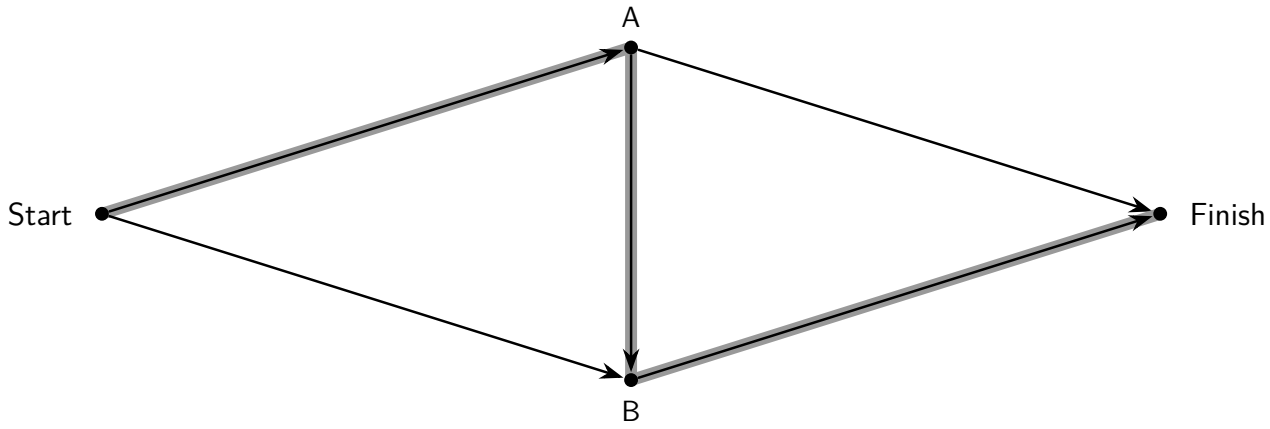


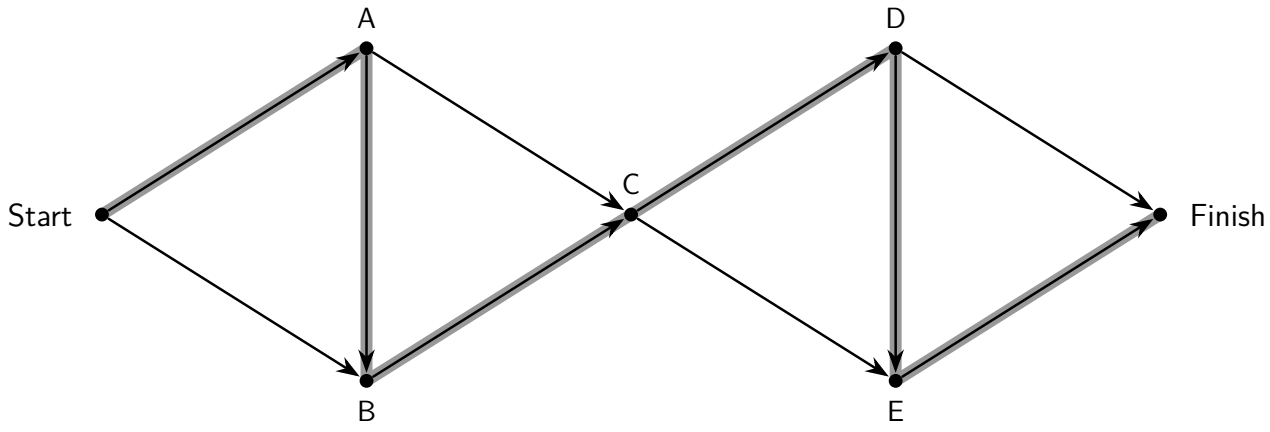
Problem 4: On each board, add one new arrow between two inside dots so that three routes fit. The new arrow cannot touch Start or Finish. Crossings without a dot are not meeting points. If it is impossible, explain why.



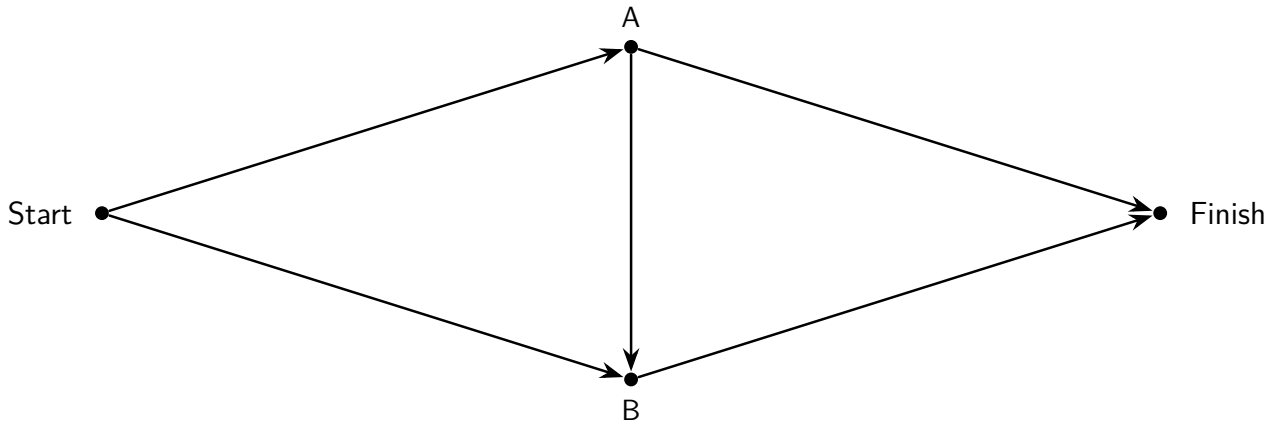


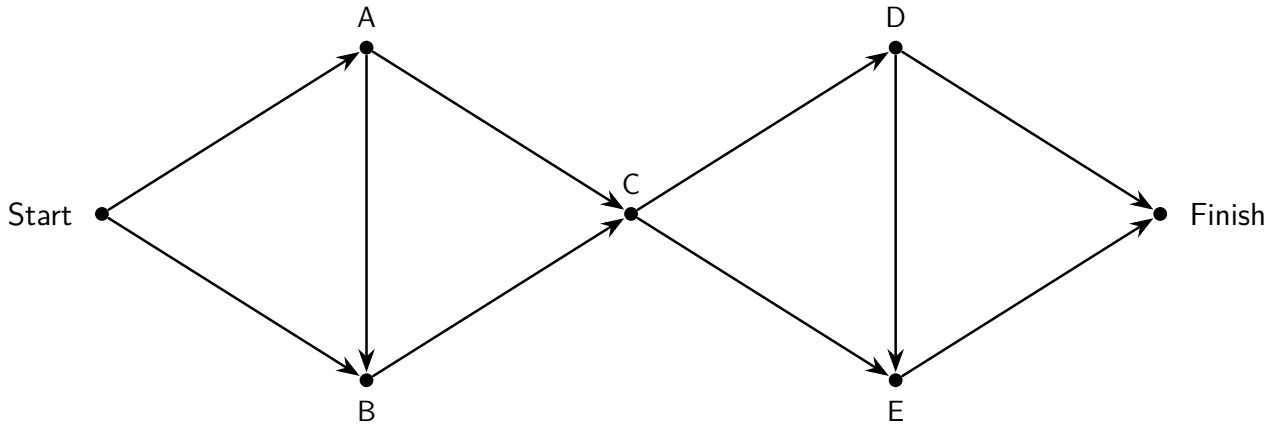
Problem 5: The thick arrows form reserved routes. No more routes fit beside them. Replace them with a largest collection on each board.





Extra workspace for Problem 5.





Problem 6: Two players take turns closing one open arrow. The player whose move leaves no route wins. Decide who can force a win on each board, and explain a strategy.

